

Smart Lab Equipment & Slot Reservation Portal

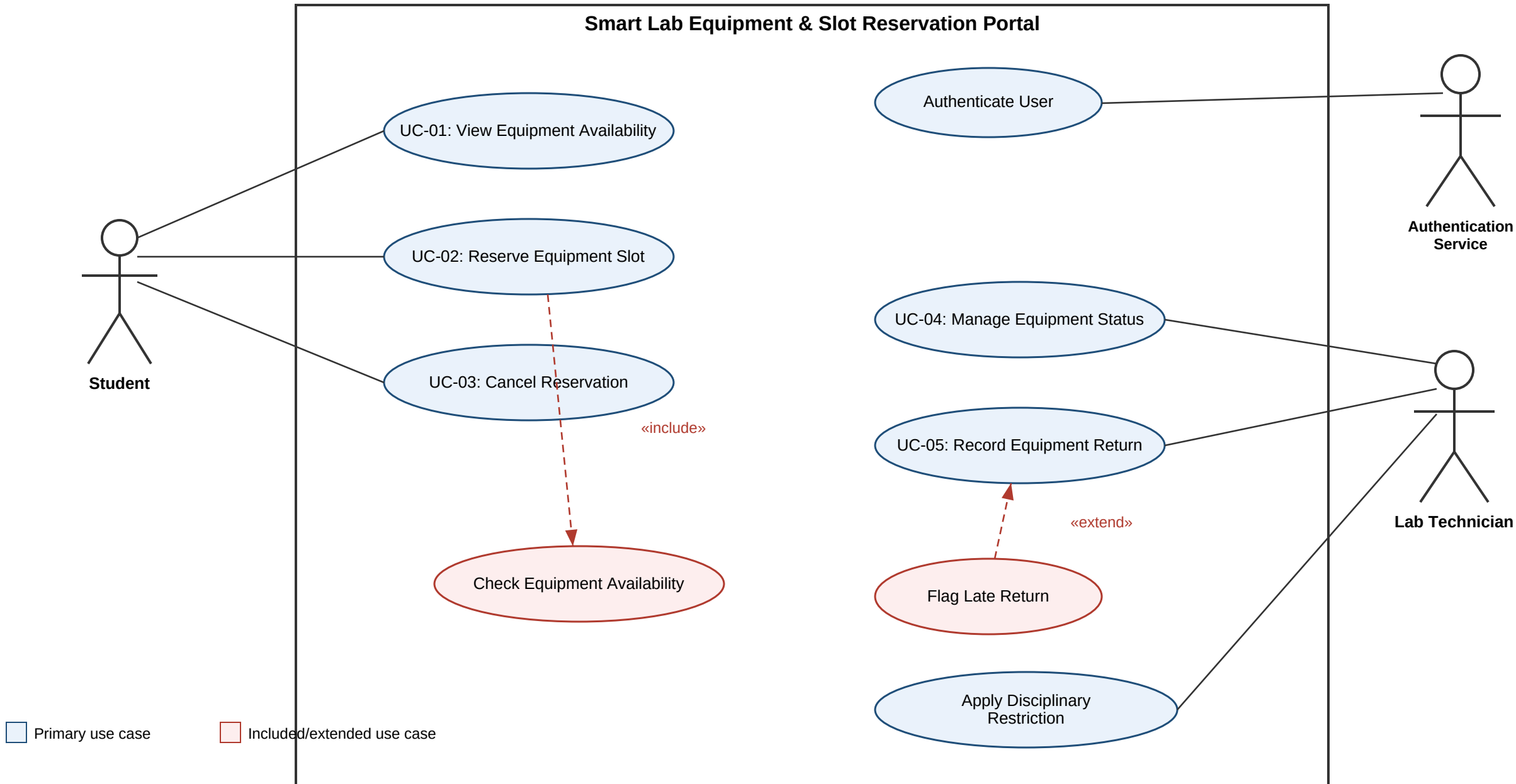
Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow authenticated students to view real-time equipment availability and reserve a maximum 2-hour slot for calibrated equipment up to 7 days in advance.	High	Pass: An available calibrated item can be reserved for 2 hours or less within the next 7 days, and a reservation token is generated. Fail: Double-booking occurs, the reservation exceeds 2 hours, or the booking is made more than 7 days in advance.	This is the core functionality of the portal.
FR-002	Functional	The system shall allow students to cancel an existing reservation before the reserved slot begins.	High	Pass: The reservation is successfully cancelled and the equipment slot becomes available. Fail: The reservation remains active after	Allows unused slots to become available to other students.

				cancellation.	
FR-003	Functional	The system shall allow the Lab Technician to update the status of equipment as calibrated, uncalibrated, under maintenance, or unavailable.	High	Pass: The updated equipment status is correctly displayed and equipment marked uncalibrated or unavailable cannot be reserved. Fail: Incorrect equipment status is displayed or unavailable equipment can be reserved.	Ensures students reserve only usable and properly maintained equipment.
FR-004	Functional	The system shall record the actual check-out and return times of reserved equipment.	Medium	Pass: Check-out and return timestamps are stored for each equipment usage session. Fail: A completed equipment usage session has missing or incorrect timestamps.	Provides accountability for valuable laboratory equipment.
FR-005	Functional	The system shall automatically flag late equipment returns and	Medium	Pass: Returns made after the reservation end time are flagged, and	Supports enforcement of late-return disciplinary rules.

		allow the Lab Technician to apply disciplinary restrictions to students with repeated late returns.		a restriction can be applied by the Lab Technician when repeated late returns occur. Fail: A late return is not flagged or a restriction is not applied according to the rule.	
NFR-001	Nonfunctional	The system shall process concurrent slot reservation requests within 200 ms with database-level lock isolation to prevent race conditions and double-booking.	High	Pass: Testing confirms the 200 ms target is met under simulated concurrent requests and no double-booking occurs. Fail: The target is exceeded or double-booking occurs.	Ensures reliable performance when multiple students attempt to reserve equipment simultaneously.
NFR-002	Nonfunctional	The system shall enforce role-based access control so that students can access only their own reservations while Lab Technicians	High	Pass: Unauthorized access attempts are rejected and each user can access only the functions permitted for their role. Fail: A user accesses	Protects reservation information and prevents unauthorized system operations.

		can access authorized equipment management functions.		another user's reservation or an unauthorized management function.	
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Use-Case Flow Specification

Smart Lab Equipment & Slot Reservation Portal

Use Case: UC-02 Reserve Equipment Slot

Primary Actor: Student

Supporting Actor: Authentication Service

Preconditions: Student has been authenticated by the Authentication Service (Authenticate User). Requested equipment exists in the system and is not marked uncalibrated, under maintenance, or unavailable (per UC-04 Manage Equipment Status).

Postconditions: A reservation record is created and locked in the database; the equipment's availability calendar is updated so the booked slot no longer appears as free to other students; a reservation confirmation is issued to the student.

Main Success Scenario

1. Student selects an equipment item (e.g., oscilloscope, logic analyzer, FPGA board) and requests to reserve a slot.
2. System performs Check Equipment Availability («include») and displays real-time availability for the next 7 days.
3. Student selects an open time slot of up to 2 hours.
4. System confirms the equipment's current status is calibrated/available (not flagged under UC-04 Manage Equipment Status).
5. System acquires a database-level lock on the requested slot to prevent concurrent double-booking.
6. System validates that the slot is ≤ 2 hours, within the 7-day booking window, and still free.
7. System creates the reservation, releases the lock, and generates a reservation token.
8. System displays a confirmation with the reservation token, equipment name, and time window to the Student.
9. Use case ends successfully.

Alternate Flow — 4a: Equipment Not Available for Reservation

- a) At step 4, if the equipment's status (set via UC-04 Manage Equipment Status) is uncalibrated, under maintenance, or unavailable, the system aborts the reservation attempt.
- b) System displays a message explaining why the equipment cannot be reserved.
- c) Student is returned to the availability view (Check Equipment Availability) to select a different item or slot.
- d) Use case ends without a reservation being created.

Alternate Flow — 6a: Slot Already Locked (Concurrent Booking Attempt)

- e) At step 6, if another request has already locked the same slot, the system rejects the current request with a “slot no longer available” message.
- f) System refreshes the availability view so the Student can pick another open slot.
- g) Use case ends without a reservation being created; no partial or duplicate reservation record is left in the database.